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S Y S T E M C O N F I G U R A T I O N

M A N U A L

P R E L I M I N A R Y

D R A F T

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1.0 Introduction

The Alarm Linker configuration utility provides the user with a tool for constructing most of the files necessary to configure a job. There are four distinct definition processes, a link process, and finally a file generation process, all of which are described in the following sections.

1.1 Index Terms

Throughout this document, the term **user** refers to the person, male or female, who is using the Alarm Linker to configure a job. The term **operator** refers to the person who will be operating the system once it has been configured and installed and is on-line.

1.2 Starting the Alarm Linker

The Alarm Linker is invoked by entering the following at the Concurrent DOS prompt:

>ALMLNK [Project Name] <cr>

The optional parameter is the one to eight character name of the project being configured (e.g. GRAHAM). This name must match the name of the MMP file generated by the VAX Sieve Compiler/Documentor. For example, if Sieve generated a file named GRAHAM.MMP, then you must enter the string 'GRAHAM' as the name of the project being configured. If you do not specify a project name when invoking the Alarm Linker, then you will be prompted to enter one subsequently.

1.3 Required Files

The following files must be present in the current directory/path in order to use the Alarm Linker:

ALMLNK.COM
ALMLNK.000
ALMLNK.001
ALMLNK.002
ALMLNK.003
ALMLNK.004
ALMLNK.005
ALMLNK.006
ALMLNK.007
ALMLNK.008

The following file is optional, but is required if the user would like help available while executing the Alarm Linker:

ALMLNK.HLP

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2.0 Build SC-300.CFG Configuration File

Selecting this option allows the user to create the system configuration file, SC-300.CFG. This file contains miscellaneous information about the project being configured, and must be present for on-line operation.

2.1 Data Entry Fields and Respective Meanings

RAM Drive
Enter the identifier of the disk drive to be used as the RAM drive. The RAM drive appears to the system like a normal disk drive, with the advantage of being many times faster. The system will load several frequently accessed files onto the RAM drive at start up time. Generally, drive 'E' is used as the RAM drive.

Display Refresh Period
Enter the time, in seconds, between successive updates of the operator's console. For example, if you enter a 10 here, the screen will only be updated every 10 seconds.

Max Recipe Data Files
Enter the maximum number of saved recipes to be allowed for each recipe class. For example, a value of 20 here would allow as many as 20 different saved recipes for every recipe class defined in the system. Given a recipe class named 'ABC', the operator could save away 20 recipes in the files named SC-ABC.001 through SC-ABC.020.

Function Key Security
Enter the initial default security level to be assigned to the operator function keys. When the system is first brought on-line, the function keys for all operator screens will be assigned the security level entered here. A value of zero corresponds to no security at all while a value of 9 yields maximum security.

Changable Points Security
Enter the initial default security level to be assigned to the changable set-points on the operator screens. Certain points may be changable on one or more of the operator screens. This security level is assigned to those points, protecting them from change by operators other than those with the proper security level. As in the function key security level above, a value of zero corresponds to no security at all while a value of nine yields maximum security.

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Typed Commands Security
There are several commands which are entered by name by the operator. The security level of these commands is set by this field. The meanings of the security levels, 0 to 9, are the same as mentioned above. In order for an operator to perform a function key, set-point change, or typed command operation, that operator must have a security level **equal to or greater** than the security level of the respective operation.

PRT Command Printer Type
Select the type of printer to be used when making hardcopy listings with the 'PRT' operator command. The list of printers supported appears as part of the user prompt for this field.

PRT Command Output Format
Select the format to be used when making hardcopies with the 'PRT' operator command. The choices are B/W graphics, color graphics, or text only.

PRT Command Output Device
Select the output destination for the 'PRT' operator command. The output can either be sent directly to the hardcopy device or can be routed to a file instead.

PRT Command Printer Port
Select the port to which the hardcopy device (printer) will be connected in the completed system. The list of options is displayed as part of the user prompt for this field. When the 'PRT' operator command is executed, the output will be sent to the port identified by this field, assuming of course the output device selected above was not a file instead.

Printer Port for Alarm/Event Printouts from the ALARM Program
Select the port to be used by the ALARM program when writing out alarm and event messages. This need not be the same as the port selected above, and in fact will most often be different. This way, one device can be logging alarms and events while a second can be used for screen dumps and so forth.

Enable Auto Logoff Feature
Using this field, the user can enable or disable automatic logoffs of operators after the operator console has been idle for a defined period of time. If the user enables auto logoff, he/she will be asked to enter the idle time which will trigger the logoff. If auto logoff is disabled, the user will not be requested to enter the idle time.

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Auto Logoff Time
This field only appears when auto logoff has been enabled as described above. Enter the time, in minutes, for which the operator console must be idle before causing an auto logoff to occur. For example, if the user enables auto logoff and enters a time of 5 (minutes), then if the operator has not made some command or data entry for five minutes, then he will be logged off.

Allow Changes if PLC not On-line
This field is generally used for debugging purposes and should be set to 'N' (no) before final installation of the system. When set to 'Y' (yes), the MMI will accept set-point changes even if no series six PLC is connected. In effect, the series six and the communication with it are 'stubbed out' when a 'Y' is entered in this field.

Base Color to Redefine
Use the arrow keys to select one of the current sixteen colors which will be replaced by another color. For example, if the user wanted everything displayed in red to be instead displayed in hot-pink, he/she would roll through the colors using the arrow keys until red was selected.

New Color
Use the arrow keys to select any one of sixty-four colors to be used instead of the color selected above. Using the above example, the user would roll through these colors until hot-pink was selected.

Print Set Point Change Events
Entering 'Y' (yes) will result in all set point changes being logged to the hardcopy device at the port defined by the **Printer**

Port for Alarm/Event Printouts from the Alarm Program field.
Entering a 'N' (no) will disable this feature.

Log Set Point Change Events
Entering a 'Y' (yes) will result in all set point changes being logged by the ALARMCON (Alarm Console) program. A 'N' will disable this feature.

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Max Samples in Circular Log Files
Enter the maximum number of samples that can be placed in a circular logging file before wrapping around and overwriting the least recent sample. As there are twenty samples per record in the file, this number must be divisible by 20. For example, if the user enters a 40 here, then 40 samples may be placed in a given logging file. When it comes time to log sample number 41, sample 1 will be overwritten and lost. Circular logging files are used to prevent the disk from becoming filled with historical logging information.

User Area for Data Logging
Enter the Concurrent DOS user area to be used for logging files. All logging files will be placed in this user area (0 to 15). By entering a 16 here, the user requests that the logging files be maintained in the user area from which the LOGGER program was run.

Enable Control Key for Touch Screen
If the system is configured with a touch screen for operator input, then this field is used. By entering a 'Y' (yes), the user is requiring the operator to depress the control key whenever the operator wishes to input data using the touch screen. In most cases this will avoid accidental input via the touch screen, but will also limit the operator-friendliness of the system. Entering a 'N' (no) allows input simply by touching the touch screen.

Slow to Fast Rate Break Point for SDCS Increment/Decrement
This field is used for the increment/decrement functions on the operator's super drive control screen(s). When an increment is requested, the slow rate is sent out to the drive. If the operator holds down the increment key longer than the time defined by this field, then the fast rate of change is sent out to the drive. The value in this field is measured in tenths of a second. For example, if the user wanted the break point to be 5 seconds, he/she would enter a 50 in this field.

2.2 Special Function Key Descriptions

Tab
This moves the cursor from the current field to the next field.

<Shift> Tab
This moves the cursor from the current field to the previous field.

F9
Function key nine exits back to the main alarm linker menu. All changes are saved.

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3.0 Define Recipe Classes

Selecting this option allows the user to define up to 32 recipe classes for the current project. This option is used only to define the existance of these recipe classes and the corresponding series six control registers associated with them. The ingredients belonging to these classes are defined using the **Edit/Create System Definition File** option on the main alarm linker menu.

3.1 Data Entry Fields and Respective Meanings

For each recipe class, the user must identify the name of that class and the series six registers for that class.

Recipe Class Name
The three character mnemonic which uniquely identifies this recipe class.

Tagname Containing Recipe Number for Requested Operation
Enter the tagname corresponding to the series six register which is used to identify the recipe number to be used in the requested operation. For example, if a recipe load is requested, the number in this register will be used to determine the recipe data file from which to get the ingredient values to be loaded. If the class name was 'ABC' and the recipe number 5, then data file SC-ABC.005 would contain the ingredient values.

Tagname Containing Recipe Number for Current Operation
Same as above, but this register contains the recipe number used in the most recently and successfully completed recipe operation.

Tagname Containing Load Request Flag
Enter the tagname corresponding to the series six register which is used to request a recipe load operation. When this register is set, the series six is requested to begin a recipe load.

Tagname Containing Save Request Flag
Enter the tagname corresponding to the series six register which is used to request a recipe save operation. When this register is set, the series six is requested to begin a recipe save.

Tagname Containing Print Report Request Flag
Enter the tagname corresponding to the series six register which is used to request a recipe print report operation. When this register is set, the series six is requested to begin a recipe report print operation. The recipe report will be created using the special report format file, SC-???.RPT, where ??? is the recipe class name. The ingredient values to be included in the report are taken from the recipe data file SC-???.NNN, which must have been created during a prior recipe save operation. NNN is the recipe number to be used in the print operation.

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Tagname Containing Print Active Report Request Flag
Enter the tagname corresponding to the series six register which is used to request a recipe print active report operation. This operation is similar to the report operation described above, except that the ingredient values are taken from the database locations corresponding to those ingredients, rather than from previously saved data files. Consequently, this operation provides a live 'snapshot' of the current recipe class.

Tagname Containing Error Status Flag
Enter the tagname corresponding to the series six register which is used to indicate the success or failure of the most recently requested recipe operation.

Tagname Containing Recipe Load Ack Flag
Enter the tagname corresponding to the series six register which is used by the series six to acknowledge receipt of a recipe load request.

Tagname Containing Recipe Save Ack Flag
Enter the tagname corresponding to the series six register which is used by the series six to acknowledge receipt of a recipe save request.

Tagname Containing Recipe Print Report Ack Flag
Enter the tagname corresponding to the series six register which is used by the series six to acknowledge receipt of a recipe print report request.

Tagname Containing Recipe Print Active Report Ack Flag
Enter the tagname corresponding to the series six register which is used by the series six to acknowledge receipt of a recipe print active report request.

Name of Related Recipe Class
Enter the three character mnemonic of the related recipe class if there is to be such a relation. This field is left blank if no related class is to be used.

3.2 Special Function Key Descriptions

Home
Displays the first of the recipe class definitions.

End
Displays the last of the recipe class definitions.

F1
Displays the previous class definition.

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F2
Displays the next recipe class definition, or a new default definition if the last definition is currently displayed.

F9
Exits back to the main alarm linker menu, saving all changes made during the current recipe class definition session.

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4.0 Edit/Create System Definition File

This option allows the user to edit the system definition file for the current project. This file must contain an entry for every tag to be used in the system. Each tag which is not a superingredient must correspond to a database location (e.g. EUR 345). Information regarding alarms, historical logging, and recipe ingredient definitions are also entered using this option.

4.1 Special Function Key Descriptions

The following special functions are available to the user from any field on any of the four definition screens.

Page Up Previous Screen
Moves the user to the previous screen for the current tag.

Page Down Next Screen
Moves the user to the next screen for the current tag. Both page up and page down work in a circular fashion. That is, paging up from screen one will move the user to screen four. In all cases, if paging would result in the user being moved to a screen which is not used for that tag (e.g. Alarm configuration screen when no alarm is configured for that tag), then the user is moved instead to the next legal screen.

Home First Tag
Moves the user to the first tag definition in the ProjectName.DEF file.

End Last Tag
Moves the user to the last tag definition in the ProjectName.DEF file.

F1 Previous Tag
Moves the user N tag definitions back, where N is defined using function key eight. For example, if N=1 and the current tag is number 5, then this function moves to tag number 4. Attempting to move back before the first tag results in a move to the first tag.

F2 Next Tag
Moves the user N tag definitions ahead, where N is defined using function key eight. For example, if N=1 and the current tag is 5, then this function moves to the definition for tag 6. If this function key is depressed and the current tag is the last definition in the ProjectName.DEF file, then a new entry will be added to the end of the file.

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F3 Copy Tag
Allows the user to copy a tag definition to the current definition. The current definition will be overwritten by the copy operation. The user must enter the name of the tag definition to copy. If the definition exists, it will be copied to the current definition. Since duplicate tagnames are not permitted, the tagname will be changed to a unique name. If a unique name can not be generated, then the copy operation will be aborted.

F4 Search / Replace Tag
This operation allows the user to search for and optionally replace a given tagname or definition field. In search only mode, the user enters a tagname to find using the wildcards '?' and '*' as required. If a tag is found, the user will be asked whether to stop the search or to find the next occurrence. If no wildcards were specified, or if the user requests that the search be terminated, then the definition of the tag which was found is read from the DEF file and displayed. If the user requests the search continue, then the next matching tagname is displayed. This continues until the user requests termination of the search or until no more tags matching the search pattern are found.

Replacement can be selected from two types, namely tagname replacement or field replacement. In the former case, the tagname found based on the search tag is replaced by that specified as the replacement tag. The latter results in the data entry field currently selected being copied into the corresponding field of the found tag. In both replacement cases, the user has the option to replace all or to be prompted for each replacement. The replacement feature can be very useful when defining tags which differ perhaps only by name or by a single field. See also the field replacement option of the block operations function described below.

F5 Delete Tag
Use this option to delete the current tag definition. A confirmation prompt will be given susequently. Once the tag has been deleted, it cannot be recovered. The delete operation can take several seconds when the ProjectName.DEF file is large.

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F6 Block Operations
There are several operations which can be performed on entire blocks of tag definitions instead of a single tag. With the exception of the **import block** operation, the user must first mark the beginning and end of the block to be operated upon before invoking one of the block operations. When the beginning and end of a block have been marked, the color of the tagnames which are within that block change. The color magenta indicates the tag is within a block marked in normal order (i.e the beginning comes before the end). However, a block can be defined in reverse order (i.e. the end comes before the beginning) in which case the tag names within that block will be displayed in the color yellow. Blocks will always be operated upon starting with the beginning of the block and finishing with the end of the block. Consequently, it is important to mark a block in the proper manner according to how the user wants the tags processed.

Mark Block Start
This operation allows the user to mark the current tag as the beginning of a block of tags.

Mark Block End
This operation allows the user to mark the current tag as the end of a block of tags.

Copy Block to Current Position
This operation copies the block of tags inserting them **before** the current tag. The name of each tag in the block that is copied is changed in order to produce a unique tag name. If a unique tag name can not be generated, that tag is not copied, although there may be other tags in the block that are copied.

Import Block
This operation allows the user to specify the name of a file from which all tag definitions will be copied into the current ProjectName.DEF file **before** the current tag. This is a very powerful tool as it allows tags from other projects to be included in the current project with very little effort on the part of the user.

Export Block
This operation copies all the tags in the current block to a file specified by the user.

Delete Block
This operation deletes all of the tag definitions in the current block. This is a particularly dangerous operation as there is no way to restore the deleted tag definitions.

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Replace Field Throughout
This operation takes the value of the field currently under the cursor and places it in the corresponding field of every tag definition in the block. This operation is similar to that in the search/replace (F4) operation.

Escape
Exit the block operations option.

F7 Insert Tag
Insert a default tag definition **before** the current tag definition. This operation may take a few seconds if there are several tags defined in the DEF file.

F8 Skip Amount
Set the number of tag definitions to skip when using the previous (F1) and next (F2) tag operations.

F9 Exit
Exit back to the main Alarm Linker menu, saving all changes.

F10 Help
Requests help for the current field under the cursor. Help is available for all fields on all four of the screens.

4.2 Main Tag Definition Screen (Screen One)

This screen contains fields for defining general aspects of a tag. This screen is always accessible for any tag in the ProjectName.DEF file.

Tag Name of Data Point
Enter the one to nine character name of the tag being defined. Only seven characters are used for the name, with the other two being used for the drive identifier (00 to FF). The default drive id is FF (255), and need not be specified explicitly. If the drive id is to be included in the tag name, then it must be the two left most characters and the name may then be up to nine characters long. However, if no drive id is to be included, then the tag name may be no longer than seven characters.

Tag Type
Enter the type of this tag, digital, analog, or superingredient. Depending on the type of the tag, certain data entry fields on screens one through four will be different.

Expanded Description
Enter the description of this tag in the space provided. The length of the description field will vary depending on the tag type.

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Engineering Units
This field only applies to analog tags. Enter the description of the engineering units in which this tag's value is measured.

Set State
This field only applies to digital tags. Enter the description of the set state of this tag. Digital tags have only two states, set and reset. For example, the set state may be ON and the reset state OFF.

Reset State
This field applies only to digital tags. Enter the description of the reset state of this tag. See the example under **Set State** above.

Historical Logging Required
Enter a 'Y' (yes) if the value of this tag is to be historically logged or if the display symbol code for this tag is 271 thru 274. Entering yes requires the user fill out the fields in screen two. A 'N' (no) prevents the user from getting to screen two.

Alarm Required
Enter a 'Y' (yes) if an alarm is to be configured on this tag. This will require the user to fill out the fields on screen three. Entering a 'N' will prevent the user from accessing screen three.

Recipe Ingredient
Enter a 'Y' (yes) if this tag is to be used as an ingredient in one or more recipe classes. This allows the user to access screen four to fill out the fields relating to ingredients. Entering a 'N' (no) prevents access to screen four.

High Data Entry Limit
This field applies only to analog type tags. If the point is going to be changable by the operator, then record the maximum value the operator shall be allowed to enter.

Low Data Entry Limit
Same as above, but for the smallest value allowed.

4.3 Historical Logging Configuration Screen (Screen Two)

This screen is used to define parameters related to the historical logging of this tag. Historical logging is used to periodically record the value of this tag for subsequent off-line analysis.

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Use Circular Logging File
Enter a 'Y' (yes) if the value of this tag is to be logged in a circular file. A circular logging file will take up to N entries, before overwriting the least recent entry with the most recent entry. Answering 'N' (no) to this question results in a linear logging file being used. Linear logging files do not overwrite themselves, but can eventually eat up all of the space on the hard disk with catastrophic results. It is best to use circular logging files unless the loss of outdated data can not be tolerated.

Initial Starting Value
This field applies to analog tags only. Enter the value to be used as the starting point whenever the SuperTrends package is used to graph the history of this tag's value.

Minimum Value Expected
This field applies only to analog tags. Enter the smallest value this tag should ever equal.

Maximum Span of Value Expected
This field applies only to analog type tags. Enter the span of values expected for this tag. The span is equal to the largest value expected less the minimum value expected.

Compression Deadband
This field applies only to analog type tags. Enter the minimum change in the tag's value required to cause the value to be logged.

Have SuperTrends Interpolate
This field applies only to those tags which are of analog type. Entering a 'Y' (yes) results in all graphs of this tag's value by the SuperTrends package being interpolated (i.e. line segments drawn between actual plots). Answering 'N' (no) results in these graphs being stairstepped in appearance.

4.4 Alarm Configuration Screen (Screen Three)

This screen is used to define the parameters associated with the alarm definition for this tag. Alarms are used to signal the operator when the values of this tag meets certain predefined criteria.

Scan Period
Enter the time, in seconds, between each check for alarm conditions on this tag. For example, entering a five here results in this tag's value being checked for alarm condition every five seconds.

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Alarm Priority
Enter the priority of this alarm. Priorities, from highest down to lowest, are Urgent, Caution, Attention, and Event.

Report Routing Zone
Enter the routing zone from one to five. The device corresponding to each zone is assigned using the **SETCOLORS** configuration tool.

Print Alarm Message
Enter the alarm message print control logic. This field determines when and if an alarm message is to be printed. The selections available differ depending on whether the tag is digital or analog.

Treat Limits as Digital
This field only applies to analog type tags. Enter 'Y' (yes) if the limits for alarm are to be digital tags. Enter 'N' (no) if the limits are constants or are analog tags.

Digital Alarm Logic
This field applies only to digital type tags. Enter an 'H' if the alarm condition is to be true when this tag is set (=1), or an 'L' if the alarm condition is to be true when this tag is reset (=0).

Enable Low Alarm
This field applies only to analog tags. If the alarm condition is to be true when the tag's value falls below the low limit, then enable the low alarm, otherwise disable the low alarm.

Enable High Alarm
This field applies only to analog type tags. If the alarm condition is to be true when the tag's value exceeds the high limit, then enable the high alarm, otherwise disable it. The Alarm Linker will force the user to enable either the high or the low alarm, otherwise there would be no need to have an alarm configured for this tag.

Low Limit
This field applies only to analog type tags. Enter the low limit for this alarm. This limit can be entered as a constant (e.g. 1234.567) or as another tag. If a tag is entered, then its type must be digital or analog according to the **Treat Limits as Digital** field above. If the value of the current tag drops below the low limit constant, or if it drops below the value of the tag being used as the low limit, then the low alarm limit condition becomes true, assuming the low limit alarm is enabled. If a tag is being used as the low limit and the tag is digital, then the low alarm limit condition becomes true if the digital tag is set.

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High Limit
This field applies only to analog type tags. Enter the high limit for this alarm. This limit can be entered as a constant (e.g. 1234.567) or as another tag. If a tag is entered, then its type must be digital or analog according to the **Treat Limits as Digital** field above. If the value of the current tag rises above the high limit constant, or if it rises above the value of the tag being used as the high alarm, then the high limit alarm condition becomes true, assuming the high limit alarm is enabled. If a tag is being used as the high limit and the tag is digital, then the high alarm limit condition becomes true if the digital tag is set.

Analog Deadband
This field applies only to analog type tags. The value entered is used as the minimum amount of change required to cause the setting or resetting of the high or low limit alarm. As an example, consider an analog deadband of 5, and a high alarm limit of 500. If the tag's value reaches 500, the high alarm condition will be true. If the tag's value then falls to 498, the high alarm will remain true. The deadband is necessary to prevent multiple setting and resetting of alarm conditions as a tag's value teeters on the edge of an alarm limit.

Alarm Action Procedure
Enter the three character name of the procedure to be invoked when the current tag is in an alarm condition. If the 'DRV' procedure is entered, the user will need to provide the drive number as explained below.

Alarm Action Logic
This field is used only when the user has entered an **Alarm Action Procedure** name which begins with the letter 'T' or 'D' and is not the name 'DRV'. The alarm action logic controls under what circumstances the alarm action procedure is invoked. The options differ depending on whether the current tag is digital or analog. For a digital tag, the options are 'D', disable invocation of alarm action procedure, 'R', invoke alarm action procedure on return from alarm condition, 'C', invoke alarm action procedure on any change of alarm condition, or 'A', invoke alarm action procedure on any alarm condition. For an analog tag, the options are 'D', disable invocation of alarm action procedure, 'H', invoke alarm action procedure on high alarm, 'L', invoke alarm action procedure on low alarm, 'E', invoke alarm action procedure on either high or low alarm, 'X', invoke alarm action procedure on return from high alarm condition, 'Y', invoke alarm action procedure on return from low alarm condition, or 'A', invoke alarm action procedure on any alarm or return from alarm condition.

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Drive Drop Number
This field is only used when the user specifies 'DRV' or 'DRG' as the alarm action procedure. Enter the number of the drive.

Associated Alarm Point
Enter the name of the tag to associate with the current tag. Using this feature, the alarm condition of the current tag can be overriden by the current alarm condition of the associated tag. The overriding condition is specified in the **Associated Point Logic** field described below. Association of alarms can be useful in reducing the number of alarm messages which are perhaps insignificant in comparison to a much more critical alarm. This field appears only if the associated point logic is not disabled.

Associated Point Logic
Enter the control logic for the associated point, or 'D' (disable) if there is to be no associated tag. Entering an 'H' will prevent the current tag from going into alarm when the associated tag is in alarm. Entering an 'L' will prevent the current tag from going into alarm when the associated tag is not in alarm.

Digital Output Point
Enter the name of the tag, if any, to be used for digital output. This tag will be set or reset, depending on the **Digital Output Logic**, when the current tag goes into alarm. This field will only appear if the user has not disabled the digital output logic.

Digital Output Logic
Enter the control logic for the digital output point. The options are 'D', disable digital output point, 'H', set digital output point when current tag goes into alarm, or 'L', reset digital output point when current tag goes into alarm.

4.5 Recipe Ingredient Configuration Screen (Screen Four)

This screen is used to define the parameters associated with the recipe ingredient definition of this tag.

Display Symbol
Enter the display symbol code which will determine how the value of this tag (ingredient) is displayed on the recipe compare function's operator screen. This field is not used for superingredients.

Display Color
Use the arrow keys to select both the foreground and background colors to be used when displaying the value of this tag (ingredient) on the recipe compare function's operator screen. This field is not used for superingredients.

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Security Level
Enter the security level of this tag (ingredient) to be used by the edit option of the recipe compare function. This field is not used for superingredients.

Drive Drop Number
Enter the drive drop number (0 to 99) to be used by the recipe system to find more information on this tag (ingredient) as regards the drive. This field is not used for superingredients.

Class Membership List
Enter the three character name of each class to contain this ingredient/superingredient.

Subingredients
This field is only used if the current tag is a superingredient. Enter the name of each tag (ingredient/superingredient) which is a subingredient of this superingredient. A superingredient can not be a subingredient of itself. A superingredient may contain up to 20 subingredients.

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5.0 Link System Definition to Project Memory Map

This option links all the defined system tagnames for this project with the MMP file for this project, which is generated by the Sieve Compiler/Documentor. All tag definitions are stored in the file ProjectName.DEF, and the memory map is stored in the file ProjectName.MMP. The user is first asked if the MMP file has changed since the last time the link operation was executed. If the MMP file has been changed, the user should answer 'Y' (yes) to this question. The MMP file will then be sorted and a separate index file created to facilitate the link operation. By answering 'N' (no), the user is indicating that the index file created previously can be used to perform the link. If no index file exists, one will be created regardless of the answer to the above question.

Next, the user is asked whether or not the link operation should pause after each error which occurs. It is generally a good idea to answer 'Y' (yes) to this question, as the user may then abort the link operation if he/she feels too many errors are being detected. By answering 'N' (no), the user prevents having the ability to abort the link operation under any circumstances.

When the link operation begins, each and every tag in the ProjectName.DEF file is searched for in the MMP index file. If the tag is found, its database prefix and offset pair is written to the ProjectName.DEF file. If the tag is not found, a tag undefined error occurs. A successful link (no errors) is necessary before generation of on-line files (Section 6.0) can be accomplished.

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6.0 Generate On-line System Definition Files

This option allows the user to generate the files necessary to run the on-line system. The user may generate all files, or select a subset of them. Obviously, all files must be generated for the on-line system to function. However, it is sometimes advantageous, especially when debugging the job configuration, to generate only a select subset of on-line files (e.g. recipe class related files). Before generating the on-line system files, the ProjectName.DEF file containing all tag definitions for the current job must have been successfully linked as described above. Like the link operation described above, the user may disable pauses after errors, although this is not always the best course of action.

6.1 Recipe Class Files

This process generates the SC-RECPE.IDX and SC-???.ING recipe class definition and ingredient files.

6.2 Tag Description Files

This process generates the ???.DES files which contain the descriptive text for each tag in the ProjectName.DEF file.

6.3 Master Ingredient/Superingredient Files

This process creates the master ingredient file (ProjectName.ING) and the Superingredient file (ProjectName.SNG).

6.4 Data Entry Limit Files

This process creates the ???LIM.IBM files for all points which are changable by the operator. For each changable point, there is an entry in one of these limit files which defines the low limit and high limit values for that point.

6.5 Alarm Configuration File

This process creates the alarm configuration file (ALARM.IBM) which contains an entry for every tag having an alarm configured on it.

6.6 Recipe Report Format Files

This process generates the recipe report format files (SC-???.RPT) from the respective un-linked format files (SC-???.ULK) created by the user with the SCREPORT configuration tool.

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6.7 Historical Logging Configuration File

This process generates the historical logging configuration file which contains an entry for each tag which either is to be historically logged or is to be displayed using display symbol code 271, 272, 273, or 274.

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7.0 Generate System Definition Reports

This option allows the user to create reports on the defined system tags or recipe classes for the current job being configured. Tab and <shift> tab are used to move between data entry fields for both search and output criteria. The currently selected field will have a light cyan label, while those not selected will be light green.

7.1 Recipe Class Reports

These reports contain information regarding the recipe classes which have been defined for the current job.

7.1.1 Search Criteria

The user may enter the three character name of a single class to include in the report, or an asterisk ('*') to include all presently defined classes.

7.1.2 Output Criteria

The user may have the report sent directly to the printer port or to a file for further processing (e.g. uploading to the VAX). If the file option is selected, the user must enter the name of the filename to receive the report.

7.2 System Tag Definition Reports

These reports contain information regarding the tags which have been defined for the current job.

7.2.1 Search Criteria

These are used to define which subset of tags are to be included in the report. When selecting an option, use the arrow keys. The option selected always appears in red.

Tagname
This field is used to select by name the tag(s) to be included in the report. Both wildcards (? and *) can be used when specifying the tags to include.

Category
The tags to include can be all, super-ingredients only, recipe ingredients only, tags which have alarms configured only, or tags set to be historically logged only.

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7.2.2 Output Criteria

These are used to determine the size and destination of the report. Use the arrow keys to select an option. The option selected appears in red.

Size
Select either a full report which contains all information on each tag or a summary which includes only the tag's name, description, and data entry limits (if applicable), and whether the tag is an ingredient/superingredient, is to be historically logged, or has an alarm configured on it.

Destination
The user may choose to have the report sent directly to the printer or alternatively to a file for further processing.

File Name
If the user has opted to have the report sent to a file, then the name of that file must be specified in this field.

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8.0 Miscellaneous Utilities

This option provides the user with seldom used but perhaps important utility programs.

8.1 DEF Index File Rebuild

Occasionally, the file containing the sorted index of the ProjectName.DEF file can become corrupted. In such a case, the user may notice, for example, that a search for a tagname instead finds the tag proceeding or following the target tag. This utility will create a new index file by sorting the ProjectName.DEF file. This does not in any way change the tag definitions or their relative ordering within the DEF file. Rebuilding the index file is a completely non-destructive operation.

8.2 DEF File Format Conversion

The need to use this utility should be extremely rare. If the user has a ProjectName.DEF file which was created with a version of the Alarm Linker **prior** to version 4.0, then that DEF file must be converted to the newer format. This **is** a destructive operation in that the old ProjectName.DEF file will be replaced by the new version. It is a good idea to back up the old version before initiating this conversion. Generally, DEF files created before 1988 will need to be converted.

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9.0 Ending an Alarm Linker Session

Select this option to exit the Alarm Linker configuration utility and return to Concurrent DOS.

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